



SGS 501/601 RESEARCH METHODS / RESEARCH METHODS and SCIENTIFIC ETHICS

STATISTICAL ANALYSIS

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AGENDA

- Collecting and summarizing data
- Probability Distributions
- Sampling distributions
- Confidence intervals
- Hypothesis testing
- Test for independence
- Regression

DEFINITIONS

Statistics: The art and science of answering questions and exploring ideas through the processes of gathering data, describing data, and making inferences about a population on the basis of a smaller sample

Population: any large collection of objects or individuals, such as citizens, students, or trees about which information is desired

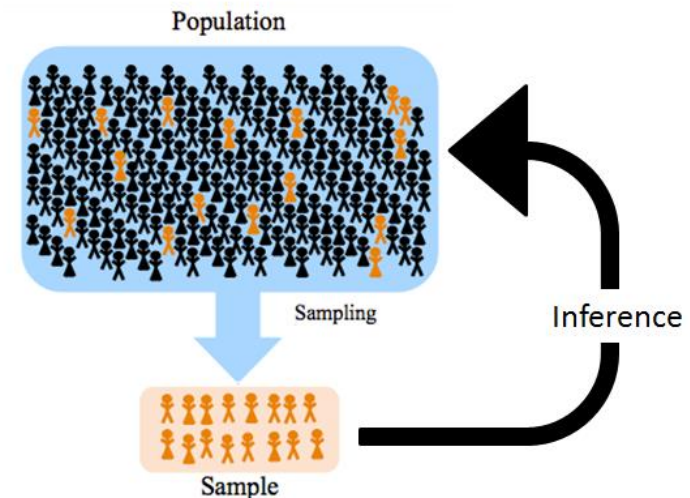
Parameter: any summary number, like an average or percentage, that describes the entire population e.g. population mean μ (mu)

Sample: is a representative group drawn from the population

Statistic: any summary number, like an average or percentage, that describes the sample, e.g. sample mean $\bar{x}$

Descriptive statistics: Techniques of describing data in ways to capture the essence of the information in the data, e.g. the sample mean

Inferential statistics: To draw conclusions from data about the population.



DATA COLLECTION

The goal of statistics is to make inferences about a population based on samples → collect sample data

Sampling based on Probability Methods

Simple random sample: making selections from a population where each subject in the population has an equal chance of being selected.

Stratified random sample: where you have first identified the population of interest, you then divide this population into strata or groups based on some characteristic (e.g. sex, geographic region), then perform simple random sample from each strata.

Cluster sample: where a random number of clusters of subjects is taken from the population of interest.

Sampling based on Non-probability Methods

- **Convenience sampling (haphazard):** Collecting data from subjects who are conveniently obtained.
- **Gathering volunteers:** Collecting data from subjects who volunteer to provide data

Types of Bias

Non-Response Bias when a large percentage of those sampled do not respond or participate

Response Bias when study participants either do not respond truthfully or give answers they feel the researcher wants to hear.

Selection bias occurs when the sample selected does not reflect the population of interest.

STUDY and VARIABLE TYPES

Observational: A study where a researcher records or observes the observations or measurements without manipulating any variables. These studies show that there may be a relationship but not necessarily a cause and effect relationship.

Experimental: A study that involves some random assignment of a treatment; researchers can draw cause and effect (or causal) conclusions. An experimental study may also be called a scientific study or an experiment.

Variable: any characteristic, number, or quantity that can be measured, counted, or observed for record.

Response Variable: Variable that about which the researcher is posing the question. May also be called the outcome or the dependent variable

Explanatory Variable: Variable that serve to explain changes in the response. They may also be called the predictor or independent variables.

Lurking (extraneous, moderating) variable

A variable that is neither the explanatory variable nor the response variable but has a relationship (e.g. may be correlated) with the response and the explanatory variable. It is not considered in the study but could influence the relationship between the variables in the study.

Confounding variable

A variable that is in the study and is related to the other study variables, thus having an effect on the relationship between these variables.

PRINCIPLES of EXPERIMENTAL DESIGN

Control: Need to control for effects due to factors other than the ones of primary interest.

Randomization: Subjects should be randomly divided into groups to avoid unintentional selection bias in the groups.

Replication: A sufficient number of subjects should be used to ensure that randomization creates groups that resemble each other closely and to increase the chances of detecting differences among the treatments when such differences actually exist.

The benefits to randomization are:

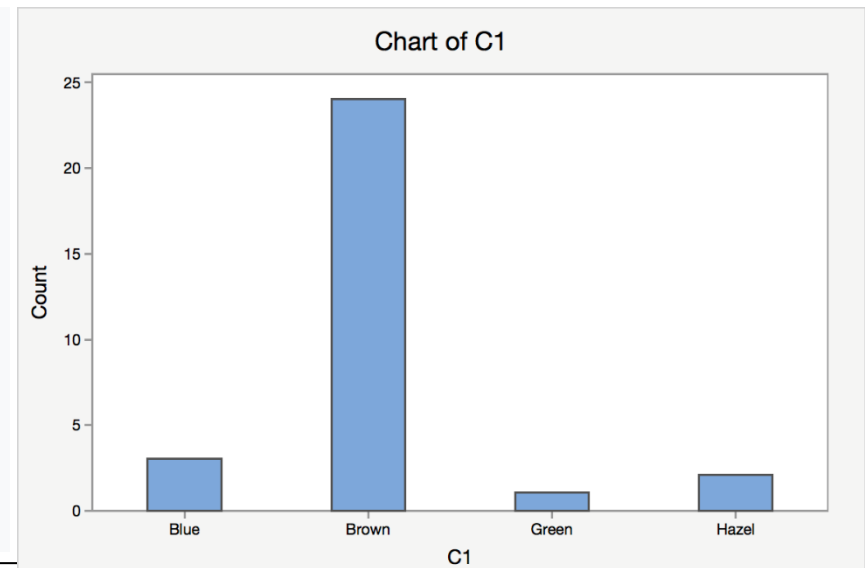
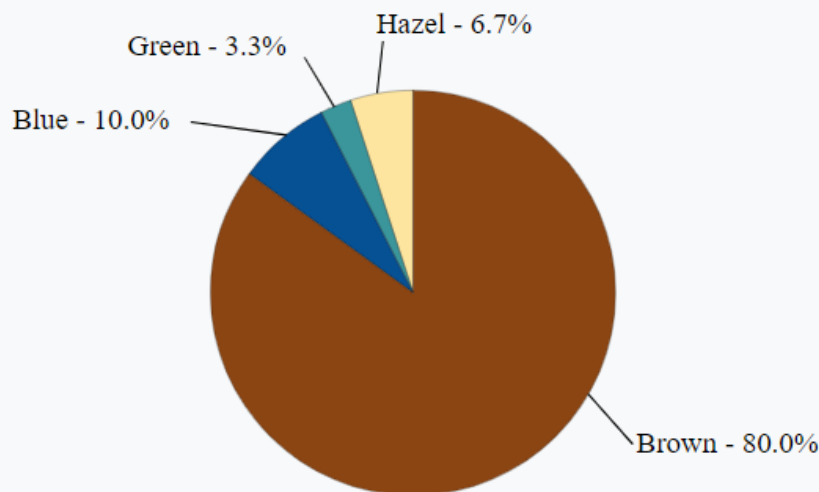
1.If a **random assignment** of treatment is done then significant results can be concluded as **causal or cause and effect** conclusions. That is, that the treatment *caused* the result. This treatment can be referred to as the **explanatory** variable and the result as the **response** variable.

2.If **random selection** is done where the subjects are randomly selected from some population, then the results can be extended to that population. The random assignment is required for an experiment. When both random assignment and selection are part of the study then we have a completely randomized experiment. Without random assignment (i.e.an observational study) then the treatment can only be referred to as being **related to** the outcome.

Summarizing One Qualitative Variable

A proportion is a fraction or part of the total that possesses a certain characteristic.
Summarize categorical data by frequencies and percentages, draw pie and bar chart

Eye Color	Frequency	Percentage
Brown	24	80%
Blue	3	10%
Hazel	2	6.6667%
Green	1	3.3333%



Summarizing One Quantitative Variable-I

Measures of central tendency: **mean**, **median** and **mode**.

Sample Mean

Let $x_1, x_2, \dots, x_n$ be our sample. The sample mean is usually denoted by $\bar{x}$

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

where n is the sample size and x_i are the measurements.

The mode is the value that occurs most often in the data. There may be more than one mode in the dataset.

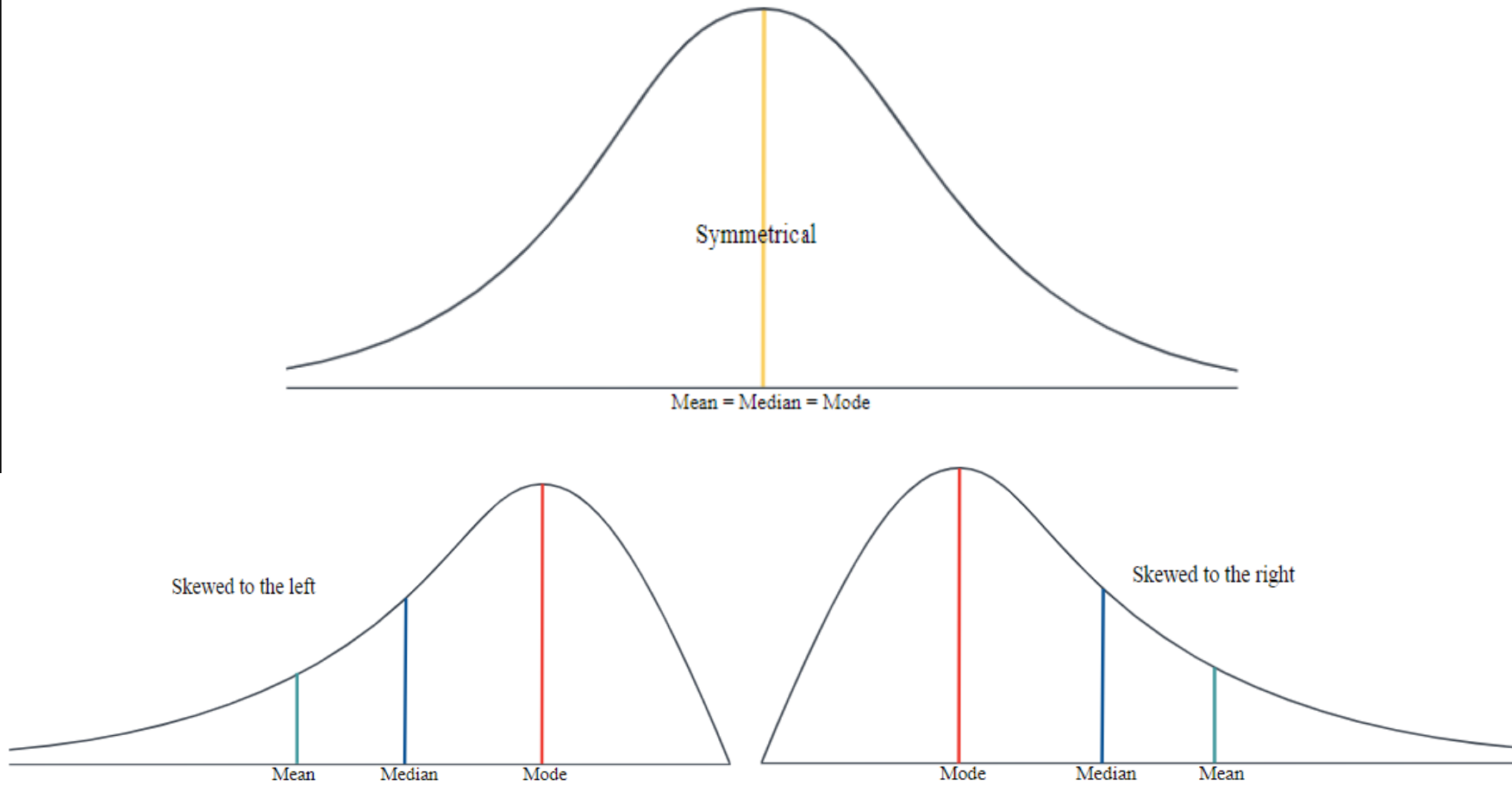
The median is the middle value of the **ordered** data. If the sample size is an odd number then the middle location point will produce a median that is an observed value. If the sample size is an even number, then take the mean of two numbers around the middle location to calculate the median.

Effects of Outliers

One shortcoming of the mean is that means are easily affected by extreme values. Measures that are not that affected by extreme values are called **resistant**. Measures that are affected by extreme values are called **sensitive**.

Summarizing One Quantitative Variable-II

Skewness: measure of the degree of asymmetry of the distribution.



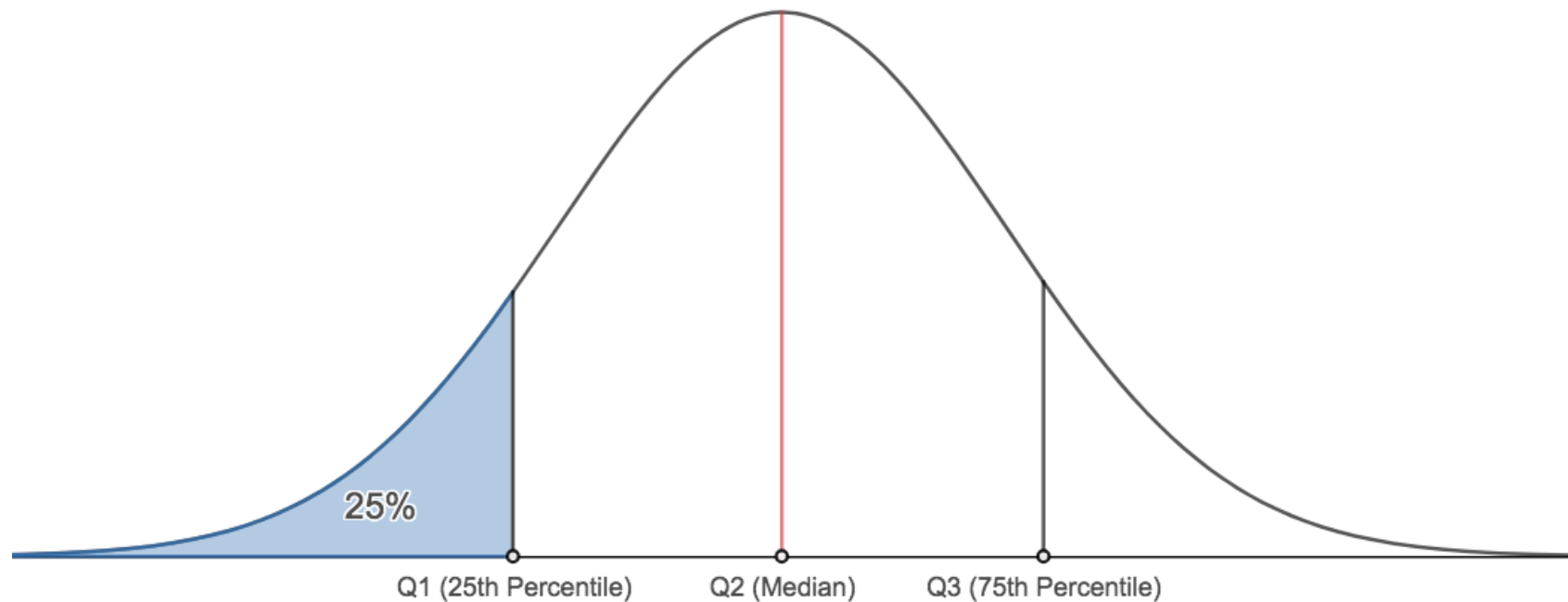
Summarizing One Quantitative Variable-III

Measures of position: range where a certain percentage of the data fall.

The p^{th} percentile of the data set is a measurement such that after the data are ordered from smallest to largest, at most, $p\%$ of the data are at or below this value and at most, $(100 - p)\%$ at or above it.

Q1: 25th percentile, **lower quartile**

Q3: 75th percentile, **upper quartile**



Summarizing One Quantitative Variable-IV

Measures of Variability or spread: Range, Interquartile range (IQR), Variance and Standard Deviation

RANGE= maximum_value – minimum_value

Interquartile Range (IQR) = Q3 – Q1

STANDARD DEVIATION (σ , s): approximately the average distance the values of a data set are from the mean

Population variance

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

where μ is the population mean and the summation is over all possible values of the population and N is the population size.

Sample Variance

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{\sum_{i=1}^n x_i^2 - n\bar{x}^2}{n - 1}$$

Where n is the sample size and $\bar{x}$ is the sample mean.

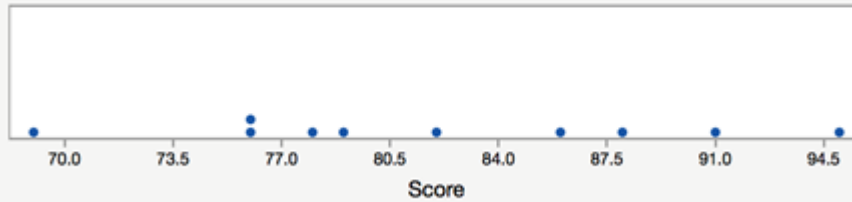
Use CV to compare dispersion (i.e. variability) of data from two or more distinct populations that have vastly different means

Coefficient of Variation (CV)

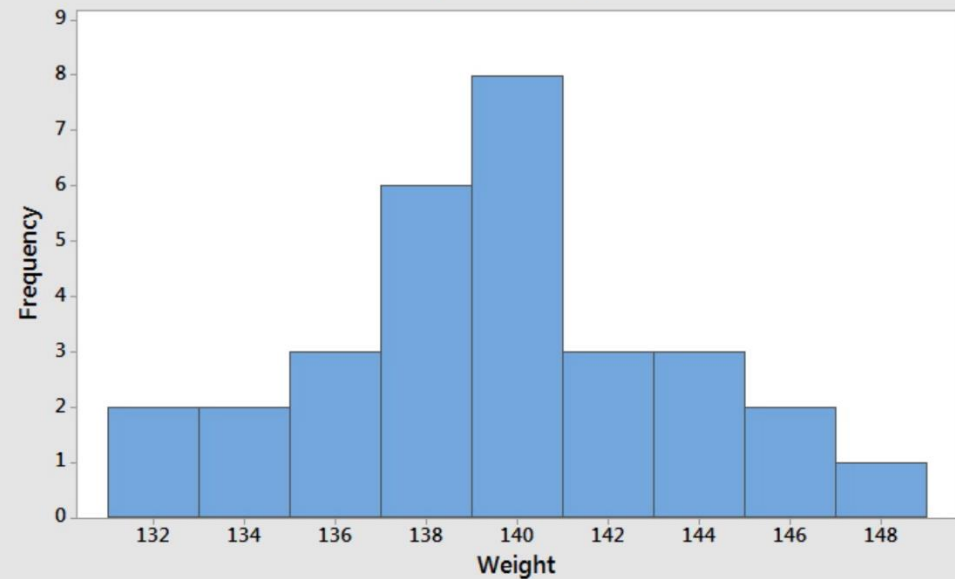
$$CV = \frac{\text{Standard Deviation}}{\text{Mean}}$$

Summarizing One Quantitative Variable-V

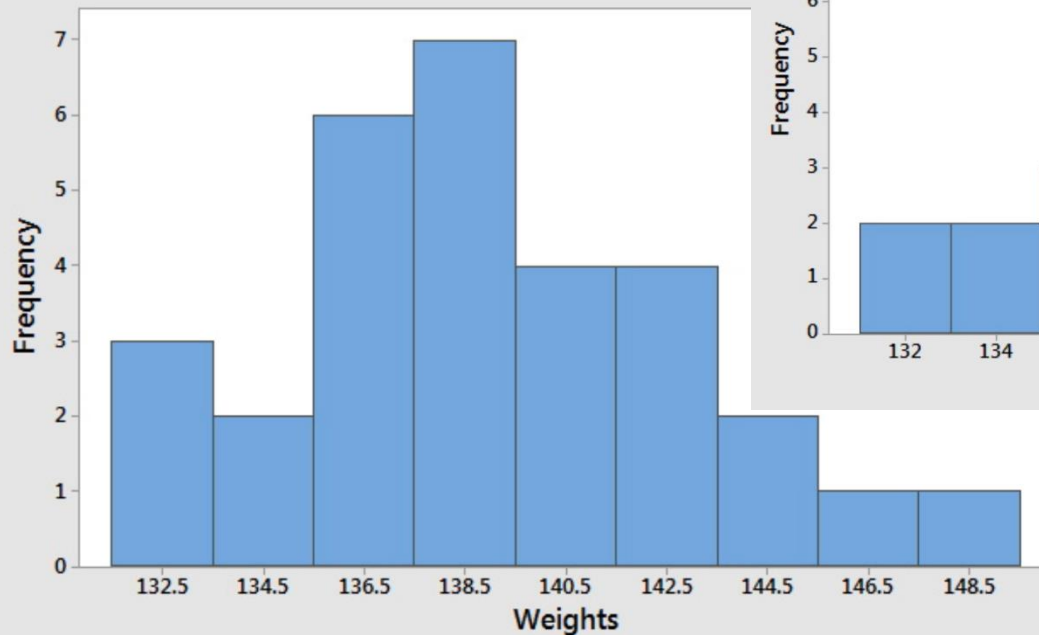
Dotplot of Score



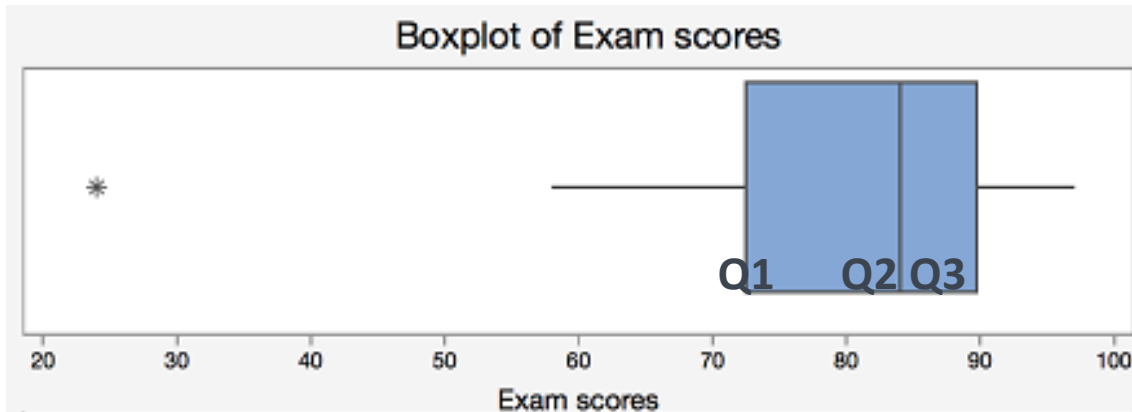
Histogram of Weight



Histogram of Weights



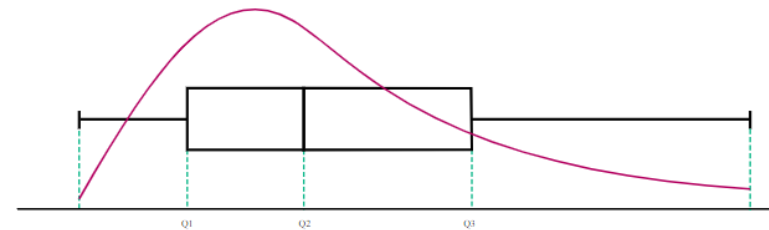
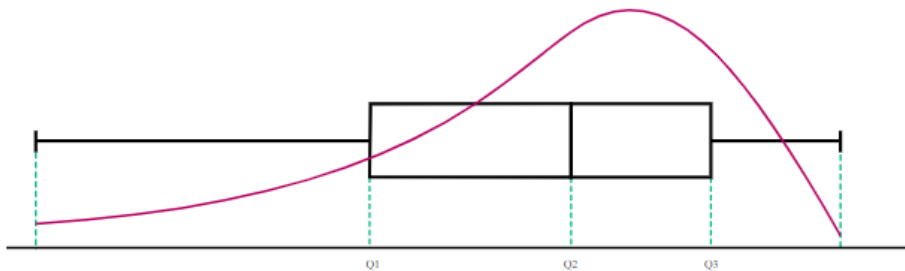
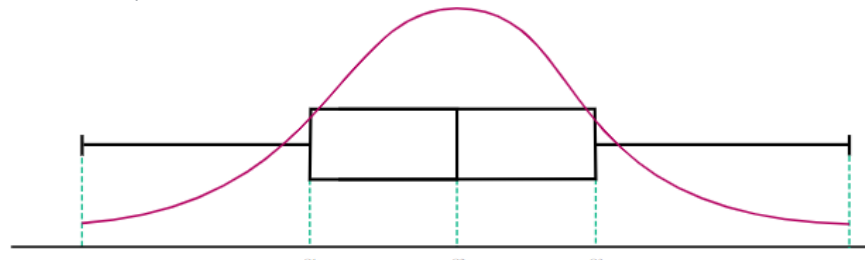
Summarizing One Quantitative Variable-VI



Potential outliers are observations that lie outside the lower and upper limits.

$$\text{Lower limit} = Q1 - 1.5 * IQR$$

$$\text{Upper limit} = Q3 + 1.5 * IQR$$



Probability

Computing Conditional Probability

The Probability of A given B:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

The Probability of B given A:

$$P(B | A) = \frac{P(B \cap A)}{P(A)}$$

Two events, A and B, are considered **independent events** if the probability of A occurring is not changed based on any knowledge of the outcome of B.

$$1. P(A \cap B) = P(A) \cdot P(B)$$

$$2. P(A|B) = P(A) \text{ or,}$$

$$3. P(B|A) = P(B)$$

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A')P(A')}$$

Probability - Exercises

Exercise-1: 10000 people have been tested for a disease where 95% of them are healthy and 5% are ill. Given that the test has an accuracy of 90%, what is the probability that a person is really ill when the test says so?

Exercise-2: Given the following table, determine whether employment status and gender are independent.

	Employed	Unemployed	Total
Male	460	40	500
Female	140	260	400
Total	600	300	900

Probability Distributions

A **random variable** (RV) is a variable that takes on different values determined by chance.

For quantitative random variables:

Discrete RV: RV can assume only a countable, sometimes infinite, number of values.

Continuous RV: RV can assume an uncountable number of values in a line interval.

A probability function is a mathematical function that provides probabilities for the possible outcomes of the random variable, X . It is typically denoted as $f(x)$.

- For discrete RV: **probability mass function (PMF):** $f(x)=P(X=x)$. PMF for a constant, x , is the probability that the random variable X is equal to x .
- For continuous RV: **probability density function (PDF):** $f(x)$ is **NOT** equal to $P(X=x)$, actually $P(X=a)=0$ but the area under the curve is equal to 1.

Cumulative distribution function (CDF), usually denoted $F(x)$, is a function that gives the probability that the random variable, X , is less than or equal to the value x . $F(x) = P(X \leq x) = P(X < x)$

Mean and Variance of Discrete Distribution

Expected value (mean) of a discrete RV:

Multiply each value, x , in the support by

its respective probability, $f(x)$, and then add them all together.

$$\mu = E(X) = \sum x_i f(x_i)$$

Variance of a discrete RV:

(Standard deviation is the

square root of variance)

$$\sigma^2 = \text{Var}(X) = \sum (x_i - \mu)^2 f(x_i)$$

$$\sigma^2 = \text{Var}(X) = \sum x_i^2 f(x_i) - E(X)^2 = \sum x_i^2 f(x_i) - \mu^2$$

Example: Binomial Distribution

1. The experiment consists of n identical trials.
2. Each trial results in one of the two outcomes, called success and failure.
3. The probability of success, denoted p , remains the same from trial to trial.
4. The n trials are independent. That is, the outcome of any trial does not affect the outcome of the others.

X = number of successes in n trials is a binomial RV

$$\mu = E(X) = np$$

(Mean)

$$\text{Var}(X) = np(1 - p)$$

(Variance)

$$\text{SD}(X) = \sqrt{np(1 - p)}, \text{ where } p \text{ is the probability of the "success."}$$

(Standard Deviation)

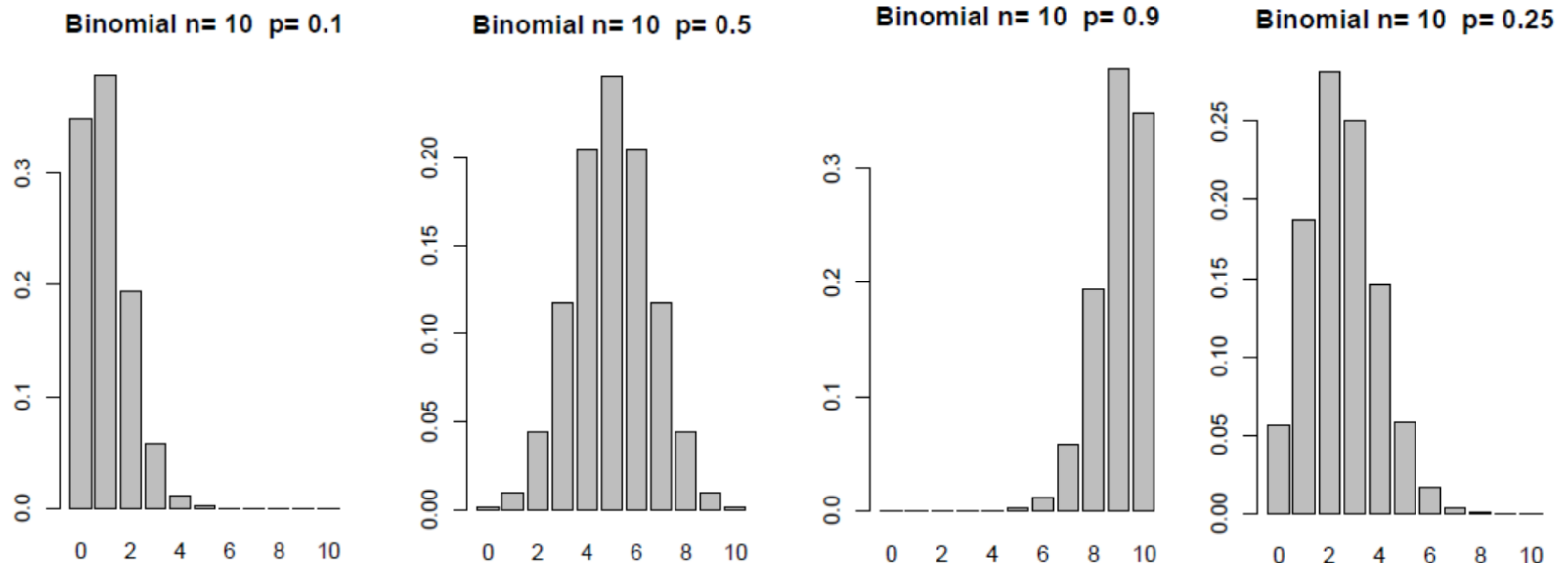
Ex: 80% of all property crimes go unsolved. Suppose that in your town 3 such crimes are committed and they are independent of each other. What is the probability that 1 of 3 of these crimes will be solved?

Graphs of Binomial Distribution

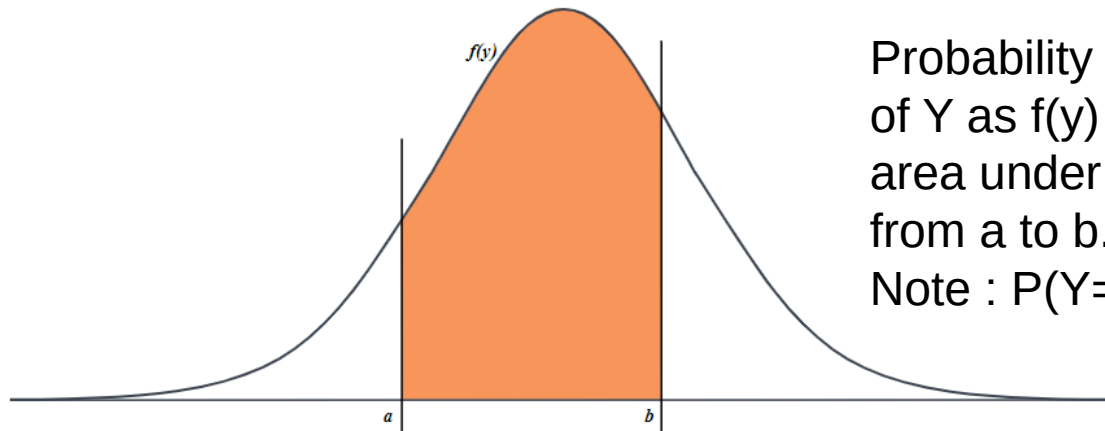
The Binomial Formula

For a binomial random variable with probability of success, p , and n trials...

$$f(x) = P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \text{ for } x = 0, 1, 2, \dots, n$$



Continuous Probability Distributions



Probability Distribution Function (PDF) of Y as $f(y)$ where: $P(a < Y < b)$ is the area under $f(y)$ over the interval from a to b .

Note : $P(Y=y)=0$

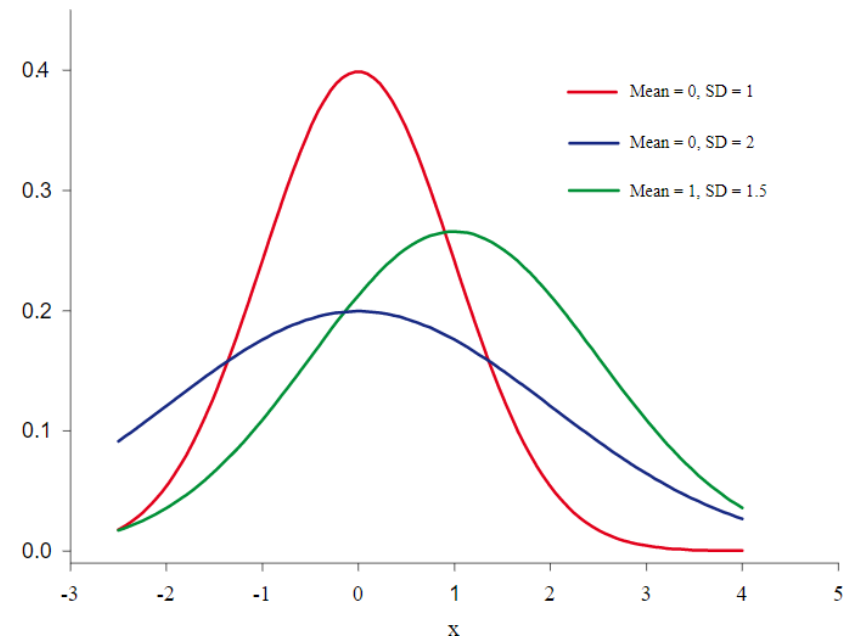
Example: Normal Distribution: $N(\mu, \sigma^2)$

- mean μ (center of the curve)
- standard deviation σ (spread about the center)

Standard normal distribution (z distribution) has a mean of 0 and variance of 1: $N(0,1)$

Any normal RV y from $N(\mu, \sigma^2)$ can be transformed to a standard normal RV by finding the **Z-score** = $(y - \mu) / \sigma$

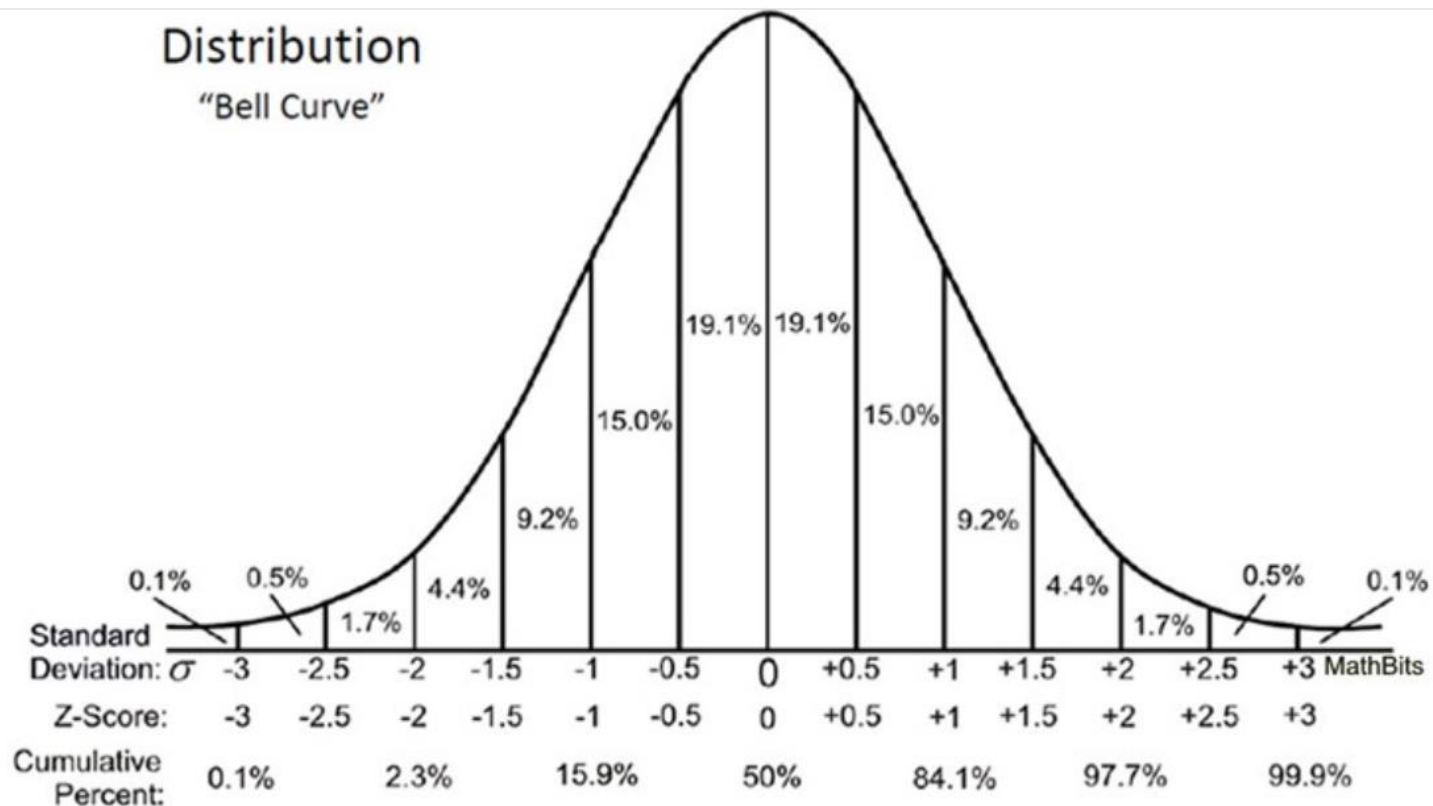
Normal Curves



Exercise: Normal Distribution

Ex-1 Assume that length of carrots is $N(14\text{cm}, 3.6\text{cm})$. What is the probability that a carrot is

- longer than 25cm,
- between 14cm and 17.6cm
- shorter than 10cm?



Sampling Distribution

The sampling distribution of a statistic is a probability distribution based on a large number of samples of size n from a given population.

Example: Choose 50 samples randomly, measure their XXX, obtain the mean. Repeat this for 1000 times (each time choose 50 samples randomly) => distribution of the sample means => expected value of this distribution will be close to the population mean

Sampling error: The error resulting from using a sample characteristic to estimate a population characteristic.

CASE-1: Population has normal distribution $N(\mu, \sigma^2)$

Sampling distribution of the sample mean is also normally distributed no matter what the sample size is. When the sampling is done with replacement or if the population size is large compared to the sample size, then $\bar{x}$ has mean μ and standard deviation $\sigma/\sqrt{n}$.

Standard Deviation of $\bar{x}$ [Standard Error]

$$SD(\bar{X}) = SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

Exercise: Sampling Distribution

The engines made by Ford for speedboats have an average power of 220 horsepower (HP) and standard deviation of 15 HP. You can assume the distribution of power follows a normal distribution.

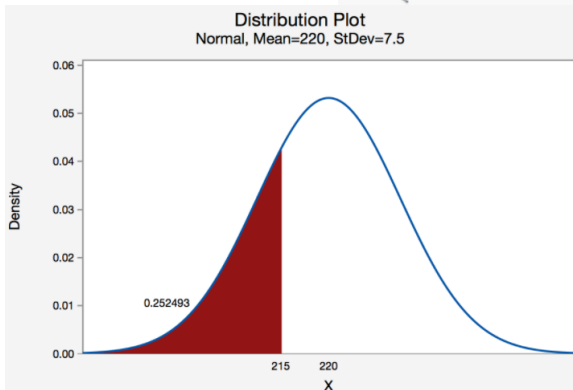
Consumer reports are testing the engines and will dispute the company's claim if the sample mean is less than 215 HP. If they take a sample of 4 engines, what is the probability the mean is less than 215?

We want to find $P(\bar{X} < 215)$.

Since the population follows a normal distribution, we can conclude that $\bar{X}$ has a normal distribution with mean 220 HP ($\mu = 220$) and a standard deviation of

$$\frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{4}} = 7.5 \text{ HP.}$$

$$P(\bar{X} < 215) = P\left(Z < \frac{215 - 220}{7.5}\right) = P(Z < -0.67) \approx 0.2514$$



If the consumer reports samples four engines, the probability that the mean is less than 215 HP is 25.14%. When the sample size is $n=100$, the probability is 0.043%.

Central Limit Theorem

CASE-2: Sample comes from a population that is not normally distributed

Central Limit Theorem

For a large sample size (sample size of $n > 30$), $\bar{x}$ is approximately normally distributed, regardless of the distribution of the population one samples from. If the population has mean μ and standard deviation σ , then $\bar{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

Ex: The weights of baby giraffes are known to have a mean of 125 pounds and a standard deviation of 15 pounds. If we obtained a random sample of 40 baby giraffes, what is the probability that the sample mean will be between 120 and 130 pounds?

We want $P(120 < \bar{X} < 130)$.

$$\begin{aligned} P(120 < \bar{X} < 130) &= P\left(\frac{120 - 125}{\frac{15}{\sqrt{40}}} < \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{130 - 125}{\frac{15}{\sqrt{40}}}\right) \\ &= P(-2.108 < Z < 2.108) \\ &= P(Z < 2.108) - P(Z < -2.108) \\ &= 0.9826 - 0.0174 \\ &= 0.9652 \end{aligned}$$

The probability that the sample mean of the 40 giraffes is between 120 and 130 lbs is 96.52%.

Confidence Intervals

REAL LIFE: We do not know the population parameters!

➔ STATISTICAL INFERENCE

ESTIMATION: Use information from the sample to estimate (or predict) the parameter of interest.

STATISTICAL (HYPOTHESIS) TEST: Use information from the sample to determine whether a certain statement about the parameter of interest is true.

POINT Estimate: An estimate for a parameter that is one numerical value. An example of a point estimate is the sample mean or the sample proportion.

INTERVAL estimates give an interval as the estimate for a parameter built around point estimates.

CONFIDENCE INTERVAL: An interval of values computed from sample data that is likely to cover the true parameter of interest.

Properties of 'Good' Estimators

- * Center of the sampling distribution for the estimate is the same as that of the population => *unbiased estimator* (commonly mean for measure of the center)
- * The estimate has the smallest standard error when compared to other estimators.

Inference for Population Mean-I

Confidence interval: sample_mean +/- margin_of_error

$$\text{Margin of error} = M \times \hat{SE}(\text{estimate})$$

*the multiplier, M , depends on our level of confidence

We are 'some level of percent confident' that the 'population of interest' is from 'lower bound to upper bound'. (Mostly 95% confidence is used!)

CASE-1: σ is known

If the population is normal with mean μ and standard deviation, σ , then the distribution of the sample mean will be Normal with mean μ and standard error $\sigma/\sqrt{n}$.

$$P\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

In other words, the $(1 - \alpha)100\%$ confidence interval for μ is:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Inference for Population Mean-II

We are 'some level of percent confident' that the 'population of interest' is from 'lower bound to upper bound'. (Mostly 95% confidence is used!)

CASE-2: σ is NOT known

If X comes from normal distribution with mean, μ , and variance, σ^2 , **OR** if $n \geq 30$, then the sampling distribution will be approximately normal with mean μ and standard error, $SE(\bar{X}) = \sigma / \sqrt{n}$

→ estimate σ by s , the standard deviation of the sample, and replace σ by s in the Z-equation. However, this new quotient no longer has a Z-distribution. Instead it has a t -distribution with $n-1$ degrees of freedom.

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

$(1-\alpha)$ 100% Confidence Interval for the Population Mean μ , (one-sample t -interval for the population mean) where the t -distribution has $df=n-1$.

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

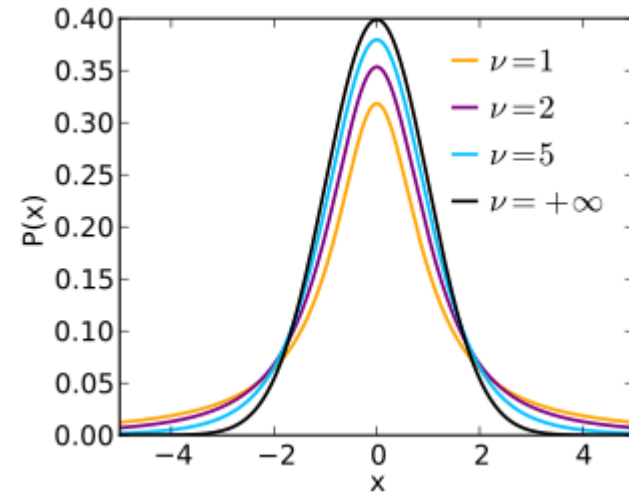
We are $(1 - \alpha)100\%$ confident that the population mean, μ , is between $\bar{x} - t_{\alpha/2} \frac{s}{\sqrt{n}}$ and $\bar{x} + t_{\alpha/2} \frac{s}{\sqrt{n}}$.

Example: Inference for Population Mean

You are interested in the average emergency room (ER) wait time at your local hospital. You take a random sample of 50 patients who visit the ER over the past week. From this sample, the mean wait time was 30 minutes and the standard deviation was 20 minutes. Find a 95% confidence interval for the average ER wait time for the hospital. Note: the standard deviation of population is NOT known.

Solution: $n=50$ the d.f. are 49, and for 95% confidence, the alpha value is 5% or 0.05. From the t-table under the column 0.025 (remember we use $\alpha/2$) and a d.f. of 40 (since 49 is not on the table), we arrive at a t-value of 2.021

If the sample size is less than 30, one needs to use a **Normal Probability Plot** to check whether the assumption that the data come from a normal distribution is valid.



$$\begin{aligned} &= \bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \\ &= 30 \pm 2.021 \frac{20}{\sqrt{50}} \\ &= 30 \pm 5.72 \\ &= (24.28, 35.72) \end{aligned}$$

Note that, $t_{0.025,49} \approx z_{0.025}$ as the degrees of freedom is 49

Normal Probability Plot

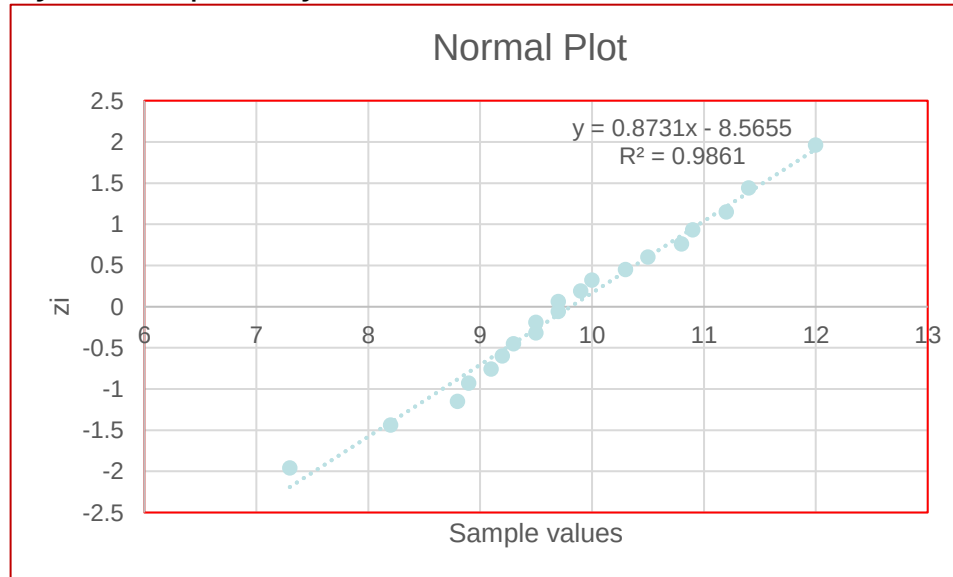
1. Order your n number of points of raw data from the minimum value to the maximum observed values.
2. Assign a rank order number (i) to each of the n points of data.
3. Calculate the cumulative probability (p_i) associated with each rank-ordered point of data. Use

$$p_i = \frac{i-0.5}{n} \quad \text{or} \quad p_i = \frac{i-3/8}{n-1/4}$$

4. Calculate the z_i value for each of your n points of data.

If $p_7 = 0.325$, you find the closest value in the body of the table and record the associated z value. For 0.325, the closest entry in the table is 0.3264, which corresponds to a z_7 of -0.45.

5. Create an x-y scatter plot of your determined z values versus measured data points.



Rank-Ordered Data	i	p_i	z_i
7.3	1	0.025	-1.96
8.2	2	0.075	-1.44
8.8	3	0.125	-1.15
8.9	4	0.175	-0.93
9.1	5	0.225	-0.76
9.2	6	0.275	-0.60
9.3	7	0.325	-0.45
9.5	8	0.375	-0.32
9.5	9	0.425	-0.19
9.7	10	0.475	-0.06
9.7	11	0.525	0.06
9.9	12	0.575	0.19
10.0	13	0.625	0.32
10.3	14	0.675	0.45
10.5	15	0.725	0.60
10.8	16	0.775	0.76
10.9	17	0.825	0.93
11.2	18	0.875	1.15
11.4	19	0.925	1.44
12.0	20	0.975	1.96

6. If the plot is “close to linear”, you may say that “the sample data can be from a normal distribution”

Hypothesis Testing

Hypothesis: a statement about a population parameter

If the sample data are consistent with the null hypothesis, then we do not reject it. If the sample data are inconsistent with the null hypothesis, but consistent with the alternative, then we reject the null hypothesis and conclude that the alternative hypothesis is true.

Six Steps for Hypothesis Tests

1. **Set up the hypotheses and check conditions: H_0 and H_a** (statement of a range of alternative values in which the parameter may fall)
2. **Decide on the significance level, α :**
3. **Calculate the test statistic** (based on sample data under the assumption the null hypothesis is true and incorporates a measure of standard error and assumptions (conditions) related to the sampling distribution)
4. **Calculate probability value (p-value), or find the rejection region:** A p-value is found by using the test statistic to calculate the probability of the sample data producing such a test statistic or one more extreme.
5. **Make a decision about the null hypothesis:** Decide to either reject the null hypothesis or decide to fail to reject the null hypothesis. Notice we do not make a decision where we will accept the null hypothesis.
6. **State an overall conclusion**

Step-1 for One-Sample Hypothesis

One Sample Mean

Research Question	Is the population mean different from μ_0 ?	Is the population mean greater than μ_0 ?	Is the population mean less than μ_0 ?
Null Hypothesis, H_0	$\mu = \mu_0$	$\mu = \mu_0$	$\mu = \mu_0$
Alternative Hypothesis, H_a	$\mu \neq \mu_0$	$\mu > \mu_0$	$\mu < \mu_0$
Type of Hypothesis Test	Two-tailed, non-directional	Right-tailed, directional	Left-tailed, directional

* μ_0 is the hypothesized population mean

Conditions: The data comes from an approximately normal distribution or the sample size is at least 30.

One-Sample Hypothesis cont.

Step 2: Decide on the significance level, α .

Typically, 5%. If α is not specified, use 5%.

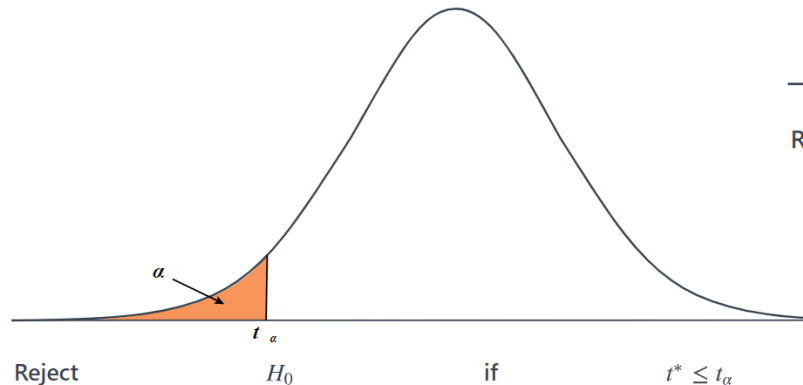
Step 3: Calculate the test statistic.

$$\text{One Mean t-test: } t^* = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

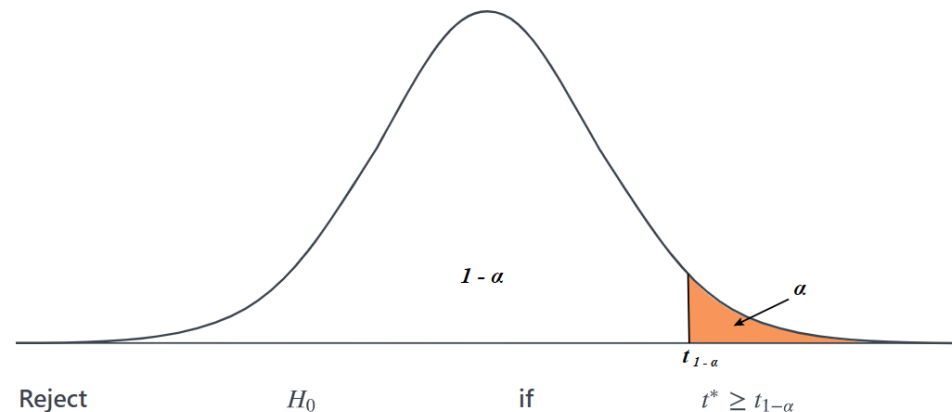
Rejection Region Approach

Step 4: Find the appropriate critical values for the tests. Write down clearly the rejection region for the problem.

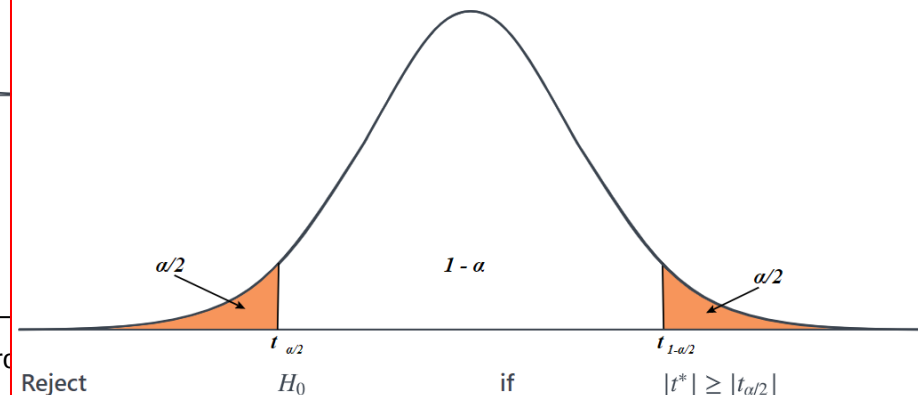
Left-Tailed Test



Right-Tailed Test



Two-Tailed Test



One-Sample Hypothesis cont.

Step 5: Make a decision about the null hypothesis.

Check to see if the value of the test statistic falls in the rejection region. If it does, then reject H_0 (and conclude H_a). If it does not fall in the rejection region, do not reject H_0 .

Step 6: State an overall conclusion.

P-Value Approach

Step 4: Compute the appropriate p-value based on our alternative hypothesis.

- If H_a is right-tailed, then the p-value is the probability the sample data produces a value equal to or greater than the observed test statistic.
- If H_a is left-tailed, then the p-value is the probability the sample data produces a value equal to or less than the observed test statistic.
- If H_a is two-tailed, then the p-value is two times the probability the sample data produces a value equal to or greater than the absolute value of the observed test statistic.

Left-Tailed

$$P(t \leq t^*)$$

Right-Tailed

$$P(t \geq t^*)$$

Two-Tailed

$$2 \times P(t \geq |t^*|)$$

Step 5: Make a decision about the null hypothesis.

If the p-value is less than the significance level, α , then reject H_0 (and conclude H_a). If it is greater than the significance level, then do not reject H_0 .

Step 6: State an overall conclusion.

Example: One-Sample Hypothesis

CASE: The mean length of the lumber is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber received has a mean length different from 8.5 feet. If the builder observes that the sample mean of 61 pieces of lumber is 8.3 feet with a sample standard deviation of 1.2 feet, what will she conclude? Conduct this test at a 1% level of significance.

Step 1: Set up the hypotheses and check conditions.

Research hypothesis is to check whether the mean is different from 8.5, we set it up as a two-tailed test:

$$H_0: \mu = 8.5 \text{ vs. } H_a: \mu \neq 8.5$$

Step 2: Decide on the significance level, α .

According to the question, $\alpha=0.01$.

Step 3: Calculate the test statistic

Rejection Region Approach

Step 4: Find the appropriate critical values for the tests.

Write down clearly the rejection region for the problem.

dof= 61-1=60, the critical value is $t_{\alpha/2}=t_{0.005}=2.660$. The rejection region for the two-tailed test is given by:

$$t^* \leq -2.660 \text{ or } t^* \geq 2.660$$

$$\begin{aligned} t^* &= \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \\ &= \frac{8.3 - 8.5}{\frac{1.2}{\sqrt{61}}} \\ &= -1.3 \end{aligned}$$

Example: One-Sample Hypothesis cont.

Step 5: Make a decision about the null hypothesis.

The observed t-value, or test statistic, is -1.3. Since t^* does not fall within the rejection region, **we fail to reject H_0 .**

Step 6: State an overall conclusion.

With a test statistic of -1.3 and critical value of ± 2.660 at a 1% level of significance, we do not have enough statistical evidence to reject the null hypothesis. We conclude that there is not enough statistical evidence that indicates that the mean length of lumber differs from 8.5 feet.

P-Value Approach

Step 4: Compute the appropriate p-value based on our alternative hypothesis.

From the t-table for 60 degrees of freedom, no value equal to 1.3, the value falls between 1.296 and 1.671. These two t-values correspond to right-tail probabilities of 0.1 and 0.05, respectively. Since 1.3 is between these two t-values, then it stands to reason that the probability to the right of 1.3 would fall between 0.05 and 0.1. Therefore, the p-value would be = $2 \times (0.05 \text{ and } 0.1)$ or from 0.1 to 0.2.

Step 5: Make a decision about the null hypothesis.

With this range of possible p-values exceeding our 1% level of significance for the test, **we fail to reject the null hypothesis.**

Step 6: State an overall conclusion.

With a test statistic of -1.3 and p-value between 0.1 to 0.2, we fail to reject the null hypothesis at a 1% level of significance since the p-value would exceed our significance level.

$$\begin{aligned}
 \text{p-value} &= 2P(T > |t^*|) \\
 &= 2P\left(T > \left| \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \right| \right) \\
 &= 2P\left(T > \left| \frac{8.3 - 8.5}{\frac{1.2}{\sqrt{61}}} \right| \right) \\
 &= 2P(T > |-1.3|) \\
 &= 2P(T > 1.3)
 \end{aligned}$$

Exercise: One-Sample Hypothesis

Emergency Room Wait Time

The administrator at your local hospital states that on weekends the average wait time for emergency room visits is 10 minutes. Based on discussions you have had with friends who have complained on how long they waited to be seen in the ER over a weekend, you dispute the administrator's claim. You decide to test your hypothesis. Over the course of a few weekends, you record the wait time for 40 randomly selected patients. The average wait time for these 40 patients is 11 minutes with a standard deviation of 3 minutes.

Do you have enough evidence to support your hypothesis that the average ER wait time exceeds 10 minutes? You opt to conduct the test at a 5% level of significance.

Exercise: solution

Emergency Room Wait Time

Step 1: Set up the hypotheses and check conditions.

At this point we want to check whether we can apply the central limit theorem.

The sample size is greater than 30, so we should be okay.

This is a right-tailed test.

$$H_0: \mu = 10 \text{ vs } H_a: \mu > 10$$

Step 2: Decide on the significance level, α .

The problem states that $\alpha = 0.05$.

Step 3: Calculate the test statistic.

$$\begin{aligned} t^* &= \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} \\ &= \frac{11 - 10}{\frac{3}{\sqrt{40}}} \\ &= 2.11 \end{aligned}$$

Step 4: Find the appropriate critical values for the tests. Write down clearly the rejection region for the problem.

The degrees of freedom for this test are $n-1=40-1=39$. The alternative is right-tailed. Therefore, we want to find the value, $t_{0.05}$, such that $P(T \geq t_{0.05}) = 0.05$.

Using the table from the text, it shows 35 and 40 degrees of freedom. We would use 35 degrees of freedom. With $\alpha=0.05$, we see a value of 1.69. The critical value is 1.69 and the rejection region is any t^* such that $t^* \geq 1.69$.

Step 5: Make a decision about the null hypothesis.

Our test statistic, 2.11, is greater than our critical value of 1.69 and therefore is in the rejection region. We would reject the null hypothesis.

Step 6. State the conclusion in words.

There is enough evidence, at a significance level of 5%, to reject the null hypothesis and conclude that the mean waiting time is greater than 10 minutes.

Comparing Two Population Parameters

Measurements taken in two variables for each sampling unit → bivariate data

CASE-1

Quantitative - taken from two distinct groups => the analysis will involve comparing two independent means, e.g.

Response variable: GPA

Explanatory variable: Graduate or Undergraduate

CASE-2

Quantitative - taken twice from each subject (paired) => the analysis will involve comparing two dependent means, e.g.

Dieting and the participant's weight before and after

Response variable: Weight

Explanatory variable: Diet

Similar way of analysis as in one-sample!

Bivariate Analysis: Theory

Theorem: Difference of two independent normal variables

Let X have a normal distribution with mean μ_x , variance σ_x^2 , and standard deviation σ_x .

Let Y have a normal distribution with mean μ_y , variance σ_y^2 , and standard deviation σ_y .

If X and Y are independent, then $X - Y$ will follow a normal distribution with mean $\mu_x - \mu_y$, variance $\sigma_x^2 + \sigma_y^2$, and standard deviation $\sqrt{\sigma_x^2 + \sigma_y^2}$.

then $X^- - Y^-$ will be approximately normal with mean $\mu_1 - \mu_2$ where μ_1 is the mean of X^- , μ_2 is the mean of Y^- .

The samples from two populations are **independent** if the samples selected from one of the populations has no relationship with the samples selected from the other population.

The samples are dependent (also called **paired data**) if each measurement in one sample is matched or paired with a particular measurement in the other sample. Another way to consider this is how many measurements are taken off of each subject. If only one measurement, then independent; if two measurements, then paired.

Inference for Independent Means-I

Compare two population means from independent populations, the difference is $\mu_1 - \mu_2$. If $\mu_1 - \mu_2 = 0$ then there is no difference between the two population parameters.

For estimating the variances for the independent samples:

- Using pooled variances (if we are reasonably sure that the two populations have nearly equal variances: neither **sample** standard deviation is more than twice the other!)
- Using unpooled (or unequal) variances

Confidence Intervals for $\mu_1 - \mu_2$: Pooled Variances

Sample-1 from Population-1: n_1, s_1

Sample-2 from Population-2: n_2, s_2

$\Rightarrow$ Common pooled standard deviation

THEN

If we can assume the populations are independent, that each population is normal or has a large sample size, and that the population variances are the same, then, t follows a t-distribution with $n_1 + n_2 - 2$ degrees of freedom.

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

$$t = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Pooled Variances

Confidence interval for the difference of two means, $\mu_1 - \mu_2$.

$(1 - \alpha)100\%$ Confidence interval for $\mu_1 - \mu_2$ for Pooled Variances

$$\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

where $t_{\alpha/2}$ comes from a t-distribution with $n_1 + n_2 - 2$ degrees of freedom.

Hypothesis test for the mean differences with pooled variances.

Null

$$H_0: \mu_1 - \mu_2 = 0$$

Alternatives

$$H_a: \mu_1 - \mu_2 \neq 0$$

$$H_a: \mu_1 - \mu_2 > 0$$

$$H_a: \mu_1 - \mu_2 < 0$$

$$t^* = \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

And t^* follows a t-distribution with degrees of freedom equal to $df = n_1 + n_2 - 2$.

Example: Inference for Independent Means

In a packing plant, a machine packs cartons with jars. It is supposed that a new machine will pack faster on the average than the machine currently used. To test that hypothesis, the times it takes each machine to pack ten cartons are recorded. Do the data provide sufficient evidence to conclude that, on the average, the new machine packs faster?

New Machine

42.1 41.3 42.4 43.2 41.8
41.0 41.8 42.8 42.3 42.7

$$\bar{x}_1 = 42.14, s_1 = 0.683$$

Old Machine

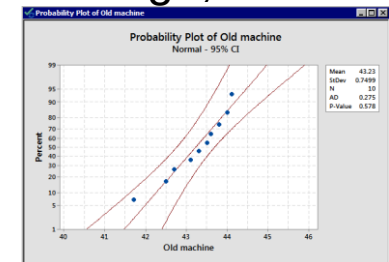
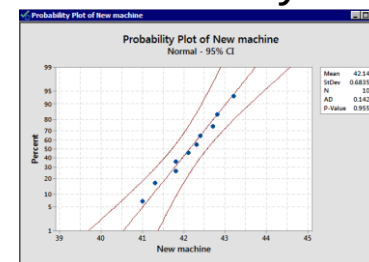
42.7 43.8 42.5 43.1 44.0
43.6 43.3 43.5 41.7 44.1

$$\bar{x}_2 = 43.23, s_2 = 0.750$$

Hypothesis Test for 5% significance using the rejection region method:

- * Are these independent samples? Yes, since the samples from the two machines are not related.
- Are these large samples or a normal population? They are not large, so, check normality!

both populations **may** come from normal Distributions!



Example cont.

Hypothesis Test for 5% significance using the rejection region method:

- Do the populations have equal variance? No information allows us to assume they are equal. We can use our rule of thumb to see if they are “close.” They are not that different as $s_1/s_2 = 0.683/0.750 = 0.91$ is quite close to 1. This assumption does not seem to be violated.

➔ proceed with the pooled t -test.

μ_1 : mean for the new machine and μ_2 : mean for the old machine.

Null hypothesis: there is no difference in the two population means, i.e.

$H_0: \mu_1 - \mu_2 = 0$

The alternative is that the new machine is faster, i.e.

$H_a: \mu_1 - \mu_2 < 0$

Calculate the pooled standard deviation:

$$\begin{aligned} s_p &= \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \\ &= \sqrt{\frac{(10 - 1)(0.683)^2 + (10 - 1)(0.750)^2}{10 + 10 - 2}} \\ &= \sqrt{\frac{9.261}{18}} \\ &= 0.7173 \end{aligned}$$

The test statistic:

$$\begin{aligned} t^* &= \frac{\bar{x}_1 - \bar{x}_2 - 0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \\ &= \frac{42.14 - 43.23}{0.7173 \sqrt{\frac{1}{10} + \frac{1}{10}}} \\ &= -3.398 \end{aligned}$$

Example cont.

Hypothesis Test for 5% significance using the rejection region method:
cont.

The alternative is left-tailed so the critical value is the value a such that $P(T < a) = 0.05$, with $10 + 10 - 2 = 18$ degrees of freedom. The critical value is -1.7341 . The rejection region is $t^* < -1.7341$.

Our test statistic, -3.3978 , is in our rejection region, therefore, we reject the null hypothesis. With a significance level of 5%, we reject the null hypothesis and conclude there is enough evidence to suggest that the new machine is faster than the old machine.

Confidence Interval:

Give a 99% confidence interval for the difference between the mean time it takes the new machine to pack ten cartons and the mean time it takes the present machine to pack ten cartons.

For a 99% confidence interval, the multiplier is $t(0.01/2)$ with degrees of freedom equal to 18. This value is 2.878. The interval becomes

$$\begin{aligned} & \bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \\ & (42.14 - 43.23) \pm 2.878(0.71173) \sqrt{\frac{1}{10} + \frac{1}{10}} \\ & -1.09 \pm 0.9232 \end{aligned}$$

The 99% confidence interval is $(-2.013, -0.167)$. We are 99% confident that the difference between the two population mean times is between -2.012 and -0.167 .

Unpooled Variances

When the assumption of equal variances is not valid, we need to use separate, or unpooled, variances.

$$t^* = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

will have a t-distribution with degrees of freedom

$$df = \frac{(n_1 - 1)(n_2 - 1)}{(n_2 - 1)C^2 + (1 - C)^2(n_1 - 1)}$$

$$\text{where } C = \frac{\frac{s_1^2}{n_1}}{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

$(1 - \alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ for Unpooled Variances

$$\bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Where $t_{\alpha/2}$ comes from the t-distribution using the degrees of freedom above.

Inference for Paired Means

What can we do when the two samples are not independent, i.e., the data is paired? Parameter of interest is the mean difference, denoted μ_d . The mean difference is the mean of the differences. We are still interested in comparing this difference to zero.

Suppose we have two paired samples of size n : $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$. Their difference can be denoted as: $d_1 = x_1 - y_1, d_2 = x_2 - y_2, \dots, d_n = x_n - y_n$. The sample mean of the differences is:

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$$

If $\bar{d}$ is normal (or the sample size is large), the sampling distribution of $\bar{d}$ is (approximately) normal with mean μ_d , standard error $\frac{\sigma_d}{\sqrt{n}}$, and estimated standard error $\frac{s_d}{\sqrt{n}}$.

$(1 - \alpha)100\%$ Confidence interval for μ_d

$$\bar{d} \pm t_{\alpha/2} \frac{s_d}{\sqrt{n}}$$

where $t_{\alpha/2}$ comes from t -distribution with $n - 1$ degrees of freedom

Example: Inference for Paired Means

Trace metals in drinking water affect the flavor and an unusually high concentration can pose a health hazard. Ten pairs of data were taken measuring zinc concentration in bottom water and surface water. Does the data suggest that the true average concentration in the bottom water is different than that of surface water? Construct a confidence interval to address this question.

Zinc concentrations

	1	2	3	4	5	6	7	8	9	10
Bottom Water	.430	.266	.567	.531	.707	.716	.651	.589	.469	.723
Surface Water	.415	.238	.390	.410	.605	.609	.632	.523	.411	.612
Difference	0.015	0.028	0.177	0.121	0.102	0.107	0.019	0.066	0.058	0.111

Response variable: concentration (quantitative)

Explanatory variable: location (bottom or surface: categorical)

The two populations (bottom or surface) are not independent => paired data

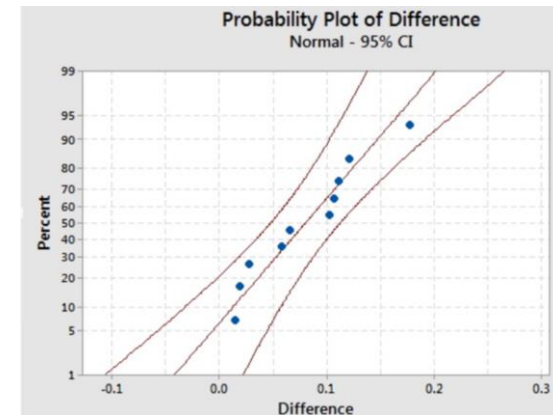
The parameter of interest is

$$\mu_d = \text{bottom cont.} - \text{surface cont.}$$

Use confidence level of 5%.

Check the conditions: no indication of normal distribution, sample size is small ($n=10$).

Use Normal Probability Plot to see if there is any violation.



Example: cont.

The sample mean difference is $\bar{d}=0.0804$ and the standard deviation is $sd=0.0523$.

With $n-1=10-1=9$ degrees of freedom, $t_{0.05/2}=2.2622$

Then the 95% confidence interval for the mean difference, μ_d is:

$$\bar{d} \pm t_{\alpha/2} \frac{s_d}{\sqrt{n}}$$
$$0.0804 \pm 2.2622 \left(\frac{0.0523}{\sqrt{10}} \right)$$
$$(0.04299, 0.11781)$$

We are 95% confident that the population mean difference of bottom water and surface water zinc concentration is between 0.04299 and 0.11781.

If there is no difference between the means of the two measures, then the mean difference will be 0. Since 0 is not in our confidence interval, then the means are statistically different (or statistically significant or statistically different).

Example Hypothesis Test for Paired Means

Does the zinc concentration data suggest that the true average concentration in the bottom water exceeds that of surface water? Conduct this test using the rejection region approach. If we consider the difference rather than the two samples, then we are back in the one-sample mean scenario. Use $\alpha=0.05$. The null and alternative hypotheses are:

$$H_0: \mu_d = 0 \text{ vs } H_a: \mu_d > 0$$

Find the critical value and the rejection region. The critical value is the value a such that $P(T > a) = 0.05$. Using the table or software, the value is 1.8331. For a right-tailed test, the rejection region is $t^* > 1.8331$

Recall: sample mean difference $\bar{d} = 0.0804$ and the sample standard deviation of the difference $s_d = 0.0523$. Therefore, the test statistic is:

$$t^* = \frac{\bar{d} - 0}{\frac{s_d}{\sqrt{n}}} = \frac{0.0804}{\frac{0.0523}{\sqrt{10}}} = 4.86$$

The value of our test statistic falls in the rejection region. Therefore, we reject the null hypothesis. With a significance level of 5%, there is enough evidence in the data to suggest that the bottom water has higher concentrations of zinc than the surface level.

Comparing Two Population Variances

To compare the variances of two quantitative variables, the hypotheses of interest are:

Null

Alternatives

$$H_0: \frac{\sigma_1^2}{\sigma_2^2} = 1 \quad H_a: \frac{\sigma_1^2}{\sigma_2^2} \neq 1 \quad H_a: \frac{\sigma_1^2}{\sigma_2^2} > 1 \quad H_a: \frac{\sigma_1^2}{\sigma_2^2} < 1$$

- F-test:** assumes the two samples come from populations that are **normally** distributed.
- Bonett's test:** assumes only that the two samples are quantitative.
- Levene's test:** assumes only that the two samples are quantitative, best to use if one or both samples are heavily skewed, and sample sizes are both under 20.

Example: **Comparing Packing Time Variances btw New and Old machines**

Descriptive Statistics				95% CI
Variable	N	StDev	Variance	for σ
New Machine	10	0.683	0.467	(0.470, 1.248)
Old Machine	10	0.750	0.562	(0.516, 1.369)

Tests				
Null hypothesis			$H_0: \sigma_1/\sigma_2=1$	
Alternative hypothesis			$H_1: \sigma_1/\sigma_2 \neq 1$	
Significance level			$\alpha=0.05$	
Method	Statistic	DF1	DF2	P-Value
F	0.83	9	9	0.787

Ratio of standard deviations	
Estimated Ratio	95% CI for Ratio using F
0.911409	(0.454, 1.829)

$s_{\text{new}}=0.683$
 $s_{\text{old}}=0.750$
 $\Rightarrow$ test statistic

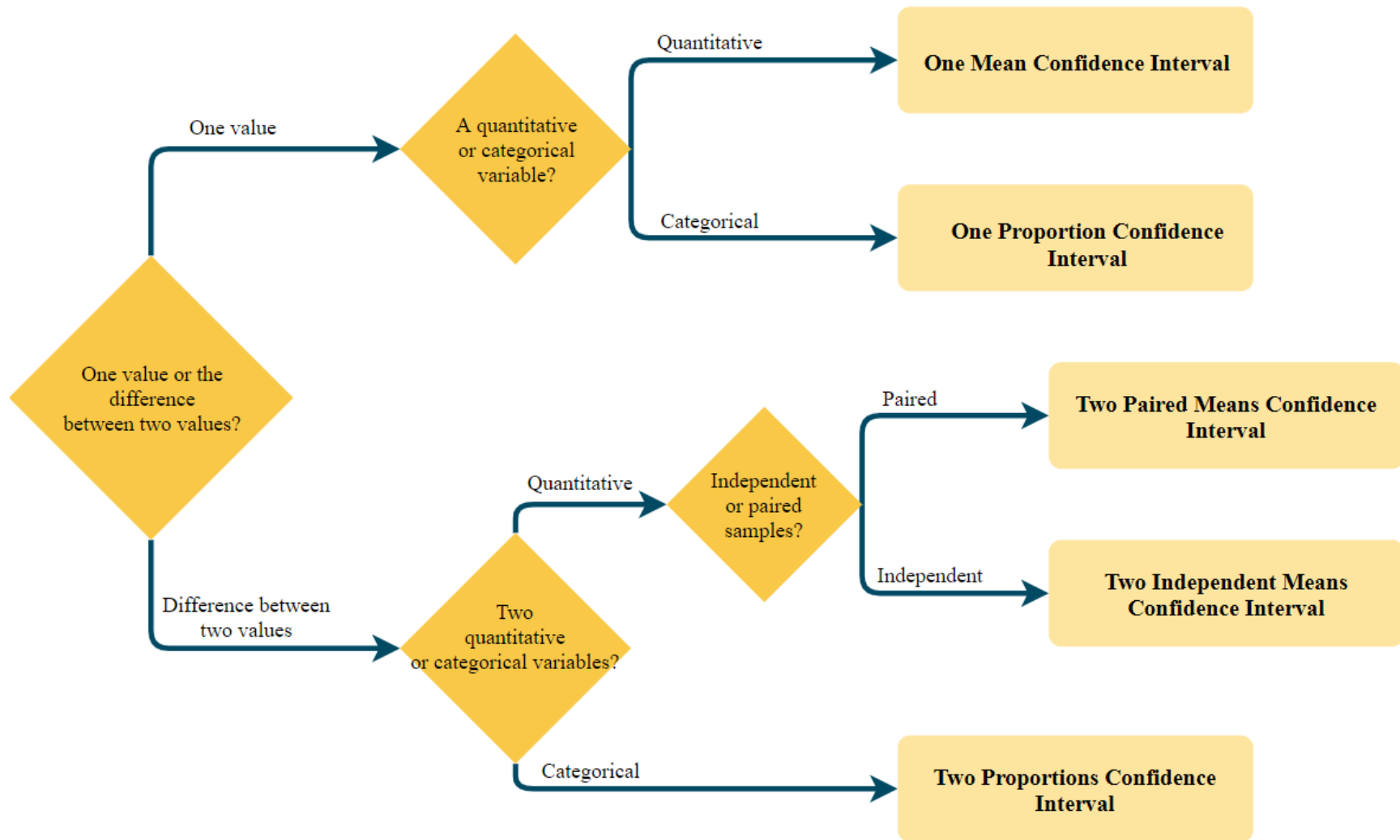
$$F = \frac{s_{\text{new}}^2}{s_{\text{old}}^2} = 0.83$$

p -value for the F -test is larger than alpha. $\Rightarrow$ we fail to reject the null hypothesis and conclude that there is not enough evidence to conclude the variances are different.

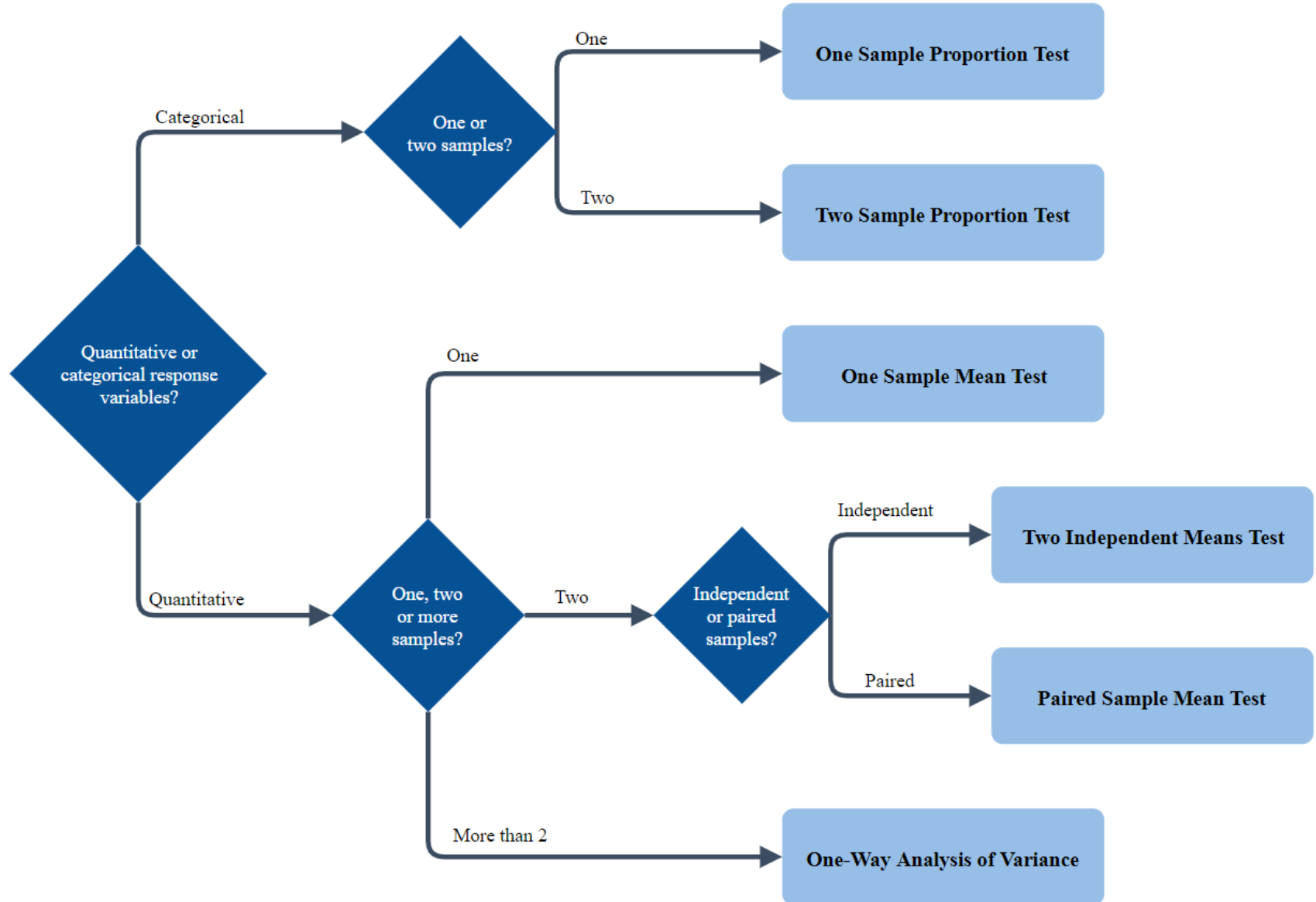
Topics not Covered

- Inference for proportions (categorical variables)
 - Chi-square Test of Independence: both variables are categorical with at least one variable with more than two levels
 - Linear Regression: both variables are quantitative
 - Analysis of Variance (ANOVA): the explanatory variable is categorical with more than two levels, and the response is quantitative
 - Nonparametric methods
- + many others...

Summary of Methods: Estimating a value



Summary of Methods: Hypothesis testing



Summary of Methods: Examine relationship

